Ronald Chiang

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Research Interests

I am fascinated by logic and its applications in theoretical computer science. I am interested in formal verification, having worked with automated reasoning tools and interactive theorem proving. More broadly, I am interested in learning about algorithms, complexity and combinatorics through the lens of logic.

Education

2026 – 2027 **Master of Philosophy in Engineering**.
University of Sydney
Topic: Best-Effort Synthesis for Quantitative Graph Games
Supervisor: Dr Sasha Rubin

2025 – 2025 **Bachelor of Science with Honours in Pure Mathematics**.
University of New South Wales, Honours Class 1, 91.750 WAM
Thesis: Ehrenfeucht-Fraïssé Games for Temporal (and) Epistemic Logics
Supervisors: Dr Paul Hunter and Dr Thomas Britz

2022 – 2024 **Bachelor of Science with Distinction in Mathematics**.
University of New South Wales, 94.087 WAM

Publications

Under Review Characterising Equivalences Between Logics Through Model Comparison Games.
Ronald Chiang, Paul Hunter. *Manuscript under review*.

Research Experience

University of New South Wales

February 26 – July 26 **Model Comparison Games**.
Created novel games for branching time, temporal epistemic and transitive closure logics. Proved original inexpressibility results for temporal epistemic logics. Provided various novel proof techniques using games.
Supervisor: Dr Paul Hunter

Sonic Labs

December 24 – February 25 **Comparing Automated Inductive Invariant Inference Tools for IVy and TLA$^+$**.
Compared the abilities of endive, for TLA$^+$, and SWISS, for IVy, to generate inductive invariants. Developed new implementation techniques for threshold and round-based consensus protocols.
Supervisors: Dr Sasha Rubin and Dr Pavle Subotić

Trustworthy Systems

February 24 – September 24 **Specifying Properties for Device-Driver Verification**.
Developed a framework to encode properties about Pancake programs in Linear Temporal Logic for HOL4.
Supervisors: Dr Johannes Pohjola and Dr Miki Tanaka

Awards

2026 – 2027 **Australian Government Research Training Program Fee Offset**.
2025 **J Holden Family Foundation Honours Award in Maths and Physics**.
2024 **Automated Proofs for TLA$^+$ – Vacation Research Internship Program Scholarship**.
2024 **UNSW Engineering Taste of Research Scholarship**.
2022 – 2024 **UNSW Science Dean's List for Academic Excellence (Top 100 in Science)**.
2023 **UNSW Engineering Dean's Award for Academic Excellence (Top 100 in Engineering)**.

Teaching

USyd Computer Science

COMP2022: Models of Computation, *26S2*.

UNSW Computer Science and Engineering

COMP1531: Software Engineering Fundamentals, *23T1, 23T2, 23T3*.
COMP2041: Software Construction, *24T1*.
COMP3121: Algorithm Design and Analysis, *23T2, 23T3, 24T1, 24T2*.
COMP3153: Algorithmic Verification, *25T3*.
COMP4141: Theory of Computation, *26T1*.
COMP4161: Advanced Topics in Software Verification, *26T3*.
COMP9020: Foundations of Computer Science, *24T2, 24T3, 25T1, 26T1*.
Course Administrator for COMP9020 in 25T1, 26T1 and COMP4161 in 26T3

UNSW Mathematics and Statistics

Mathematics Drop-In Centre, *24T1, 24T2, 24T3, 26T1, 26T2*.
Mathematics Exam Invigilation, *26T1, 26T2*.
MATH1011: Fundamentals of Mathematics, *26T1*.
MATH1081: Discrete Mathematics, *25T2, 25T3*.
MATH1131: Mathematics 1A, *26T1*.
MATH1151: Mathematics for Actuarial Studies and Finance 1A, *26T1*.
MATH1231: Mathematics 1B, *25T2, 26T1, 26T2*.

Extracurriculars

2024 **Vice President of Education**, *UNSW Mathematics Society*.
2023 **Director of Outreach**, *UNSW Mathematics Society*.
2022 **Subcommittee Member of Education**, *UNSW Mathematics Society*.
2022 – 2023 **AV Simulations Developer**, *UNSW Redback Racing*.

Referees

Sasha Rubin, *University of Sydney*, sasha.rubin@sydney.edu.au.
Paul Hunter, *University of New South Wales*, paul.hunter@unsw.edu.au.
Thomas Britz, *University of New South Wales*, britz@unsw.edu.au.